Does Rate of Return (ROR) on Education Differ for Men and Women?

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Abstract: This empirical economics study examines gender-based disparities in the Rate of Return (ROR) on education in the United States labor market, utilizing 2023 IPUMS CPS data. Significant findings emerged after implementing OLS regression with optimal model selection, including an interaction term between male and years of education, in addition to other control variables. While men generally experience around 28.9% higher hourly wage compared to women if we only consider gender and hold all other control variables constant; notably, the wage gap narrows by approximately 1.46% with each additional year of education. Particularly, males and females with equivalent educational attainment are likely to earn equal hourly wages after two decades of education. These results demonstrate unexpected shifts in earning differentials, underline the multifaceted dynamics that contribute to income disparities, and emphasize the intricate relationship between education, gender, race, and citizenship status in the US labor market.

I. INTRODUCTION

In today's evolving landscape of the US labor market, the question of whether the Rate of Return (ROR) on education differs between genders has been a crucial concern for economists and policy makers. This inquiry involves not only genders but also multiple factors that influence hourly wages. In this paper, I utilize Integrated Public Use Microdata Series (IPUMS) Current Population Survey (CPS) dataset for the year 2023, March ASEC, and run several Ordinary Least Squares (OLS) regression models to find an optimal model that can well present the data, seeking answers for the research question: “Does Rate of Return (ROR) on Education Differ for Men and Women?” In this model, my dependent variable, hourly wage, is transformed using natural logarithm to discern percentage differences accurately.

I have read several other research papers investigating the same relationship, but I want to expand further, not only by including multiple independent variables, but also by creating a dummy variable indicating male and including an interaction term to combine education years and male to directly examine whether the relationship between the ROR on education and male changes depending on the value of education levels.

My choice to explore the Rate of Return (ROR) on education across genders lies on both academic and personal interests. Initially, I was inspired by Human Capital Theory, which suggests that investing in education should correlate with increased earnings. However, I believed that there existed so many other factors determining a person' wage, not only educational attainment alone. This paper discovers the complexities of real-world scenarios, considering factors like gender-based wage gaps and discrimination against marginalized groups like females, Black individuals, and non-US citizens. The urgency of addressing persistent gender disparities in the job market amplifies the importance of this investigation, aiming to contribute empirical insights to discussions on gender equality and fair wages. Moreover, my personal experiences as a female student having an Asian background and pursuing a bachelor's degree in the STEM field also urges me to explore potential biases and challenges that may impact my future earnings and career trajectories.

The remainder of my paper will unfold in a structured manner, starting with a comprehensive literature review that synthesizes Human Capital Theories and previous empirical studies on gender-based wage differentials, setting the stage for my analysis. Following this, I will describe the dataset and variables used, providing a clear understanding of the empirical foundation of my study. The most important part of my analysis lies in presenting and interpreting five regression models, progressing from a basic model focusing solely on education

years to the most optimal model with natural logarithm of hourly wage regressed on all independent variables, including the interaction term between gender and education years, four race dummies, and eleven occupation dummies to see the percentage differences. Each regression model will be thoroughly analyzed to uncover insights into the complexities of wage determination, shedding light on factors that transcend traditional explanations.

II. LITERATURE REVIEW

The connection between educational attainment and income levels has been a central focus in empirical economics, particularly within the context of the United States labor market. However, researchers always want to find out if educational attainment is enough, or if there is any bias based on physical traits, especially gender, that we cannot control and are randomly assigned by luck since we were born. Understanding how educational achievements translate into economic outcomes, and the mediating influence of gender, is crucial for addressing income disparities and promoting equitable opportunities. There are a number of studies that look at the effects of education on earnings different between males and females. This literature review synthesizes recent empirical studies to explore the causal relationship between educational attainment and income levels, and how gender mediates this relationship.

According to Sorel and Shinners (2017), Human Capital Theory serves as a foundational framework for understanding the relationship between education and income. For each increase in education year, the wages will rise by 12.6% (Sorel et al., 2017). As stated by this theory, investments in education lead to the accumulation of skills, knowledge, and abilities that enhance individuals' productivity, thereby increasing their earning potential. However, this relationship is complex and influenced by various factors, including labor market dynamics, discrimination, and

individual characteristics. Understanding that, they use information extracted from the 2017 American Community Survey Public Use Microdata Sample specific to the state of Georgia to examine further and find out that educational attainment did not emerge as the primary determinant of wages. Surprisingly, they find out that gender had a more substantial impact on wages within the context of their project. Significantly, their model indicates that women's salaries are 48.7% lower when holding all factors constant.

Gender disparities in income have been a persistent issue, despite advancements in educational attainment among women. This has been known as "Gender Wage Gap," which means women tend to earn less than men. Economists like Bertrand, Goldin, and Katz (2010) have extensively discovered this problem. According to their definition, various societal and structural factors come into play, such as occupational segregation, discrimination, and caregiving responsibilities, all of which contribute significantly to the disparity in earnings between genders. Their research emphasizes how differences in career paths and individual preferences, influenced by societal norms and expectations, ultimately lead to unequal pay outcomes. "A man in the ninetieth percentile earns over $1 million at 10 to 16 years out as compared to $438K for the ninetieth percentile women" (Bertrand et al., 2010).

Another recent finding from Pew Research Center indicates that in 2022, females in America earned around 82 cents for each dollar earned by males, resulting in an estimated gender wage gap of 18% (Kochhar, 2023). They consider factors like discrimination, occupational segregation, and societal norms, all of which contribute to the gender-based wage gap.

Moreover, not just in the United States, it is informative to expand the scope and see this earning differentials on a global scale. Rigobon and Hogan (2002) apply their methodology to

analyze data from the Labour Force Survey in the United Kingdom, revealing significant insights into the returns to education during the sample period. Specifically, their study indicates that the return to education for men during this period was approximately 7.7%. This finding underscores the importance of education in shaping wage differentials and highlights the potential impact of educational attainment on economic outcomes within labor markets. While this study focuses on the UK context, its implications resonate with broader discussions surrounding human capital investment and its implications for wage structures, providing a valuable reference point for understanding similar dynamics in the US labor market.

Besides education and genders, the impact of occupational choices on income distribution has been a focal point of extensive research, particularly within the context of the United States. In their groundbreaking work, He and Jiang (2023) provide valuable insights by meticulously dissecting income inequality and shedding light on the persistent occupational disparities in income distribution. Their rigorous analysis draws from data sourced from the March Current Population Survey (CPS), allowing them to delve into the intricate dynamics of escalating income inequality across the nation. Their investigation meticulously dissects income inequality into components linked to five crucial individual traits: sex, race, education, occupation, and industry of work. They demonstrate that while education and gender do play a role in narrowing the gap between top and bottom earners to some extent, it is essential to recognize that occupational disparities persist. These disparities significantly impact the returns on education for different groups (He et al., 2023). Their paper is based on the theory that skill-biased technological change has exacerbated income inequality by increasing the demand for highly skilled labor, leading to a significant earnings gap between workers with specialized skills and those without. This discovery holds immense significance for my research, as it provides a

compelling basis for a more comprehensive examination of how occupational bias contributes to the existing income gap. Therefore, I include the occupation variable in my regression model, dividing occupations into eleven groups as dummy variables to extend my answer for the research question. Notably, their findings reveal that shifts in the returns to education serve as a substantial driving force behind the observed increase in inequality.

The review of literature provides a comprehensive understanding of how educational attainment influences income levels and how these are different between people in different groups, especially individuals in different genders, within the United States context. Drawing insights from referenced papers and their findings and compelling numbers from varied years, I leverage Stata software to conduct regression analysis using current data from the Integrated IPUMS CPS, March ASEC 2023. My paper seeks to contribute evidence-based policy recommendations that address income disparities and promote economic mobility, aiming to uncover the differences in Rate of Return (ROR) on education between males and females, while also considering other factors such as race, occupation, experience levels, and citizenship status.

III. THEORETICAL ANALYSIS

Human Capital Theory, pioneered by Becker (1964) and further developed by economists like Mincer (1974), suggests that investing in education enhances an individual's productive capacity, subsequently leading to higher earnings. This theory serves as the foundation for my analysis, guiding me to investigate how educational attainment impacts income levels and if this impact differs between men and women.

Complementing this is the Gender Wage Gap that acknowledges the persistent disparities in earnings between men and women, indicating that women, on average, tend to earn less than men with similar educational backgrounds.

To translate the above Human Capital Theory and Gender Wage Gap issue into empirical analysis, I apply economics theoretical foundation for some special terms in my optimal regression model. Firstly, I use the natural logarithm of hourly wage and regresses it on education years, male, interaction term between education years and male, and several other independent variables. The transformation from hourly wage to its logarithm is common in economics as it allows us to interpret results in percentage terms, providing a clearer picture of wage differentials.

Secondly, the inclusion of an interaction term between education years and male gender is crucial as it helps me understand how the impact of education on wages varies between men and women, capturing potential gender-based disparities in the Rate of Return (ROR) on education and if this gap changes as the education attainment increases.

Additionally, using experience squared besides experience alone in my regression model allows me to capture potential nonlinear effects and variations in the relationship between experience and wages over time, leading to more accurate and robust statistical analysis and interpretation.

Most importantly, I use Ordinary Least Squares (OLS) regression for my analysis to answer the research question because OLS is widely used for estimating the relationship between a continuous dependent variable, hourly wage in this context, and multiple independent variables like education years, gender, experience, and other factors. OLS regression offers several advantages including linearity assumptions, efficient coefficient estimates, interpretability of

results, statistical inference capabilities for hypothesis testing and confidence intervals, and the ability to check and address model assumptions.

Overall, this theoretical framework and model specification enable me to delve into the nuanced dynamics of the ROR on education across genders, considering a range of factors that influence wage outcomes and contributing to a deeper understanding of income inequalities in the labor market.

IV. EMPIRICAL ANALYSIS

A. The Data

The data for this paper was collected from the Current Population Survey (CPS), specifically the March Annual Social and Economic Supplement (ASEC) 2023 and used for regression in Stata. ASEC data is conducted annually and is advantageous because it provides a comprehensive and nationally representative sample of households and individuals in the United States. The March ASEC 2023 dataset includes detailed information on demographics, employment status, income, education, and other relevant variables, making it well-suited for analyzing labor market dynamics, income differentials, and the impact of education on earnings with a high degree of accuracy and reliability.

To conduct analysis and answer the research question for my paper, I used one dependent variable: hourly wage (HOURWAGE), the amount of money paid to an employee for each hour of work performed, typically in a job that pays on an hourly basis rather than a fixed salary. I sorted this data and excluded all errors and outliers that displayed as 999.99 (Not in Universe). Initially, the original dataset comprises 146,133 observations of the variable hourly wages (HOURWAGE). However, after the process of removing outliers, the sample size was reduced to

3,795 observations. This reduction was necessary to prevent extreme values from unduly influencing the results. However, it is also important to note that this smaller sample size may limit the generalizability of my findings and could impact the statistical power of the analysis.

About independent variables, I have education years (EDUC) and sex (SEX) as my two main considerations. However, I added race (RACE), citizenship status (CITIZEN), occupation (OCC), hours usually worked per week (UHRSWORKT), and experience to enhance the layers of complexity to my analysis. All these variables are divided into 2 groups: Continuous variables and Dummy variables.

a. Continuous variables

Firstly, for education years, I re-coded the dataset to remove all people with no schooling and make the number of education years ranging from 9 (grade 9th and lower) to 18 (master degree and higher). This new variable for education years is called "educyears".

For variable hours worked per week, which captures the intensity of labor input and allows for the examination of how varying levels of work hours impact earnings and economic outcomes, I re-coded "uhrsworkt" variable on Stata to remove all values that were more than 168, since there are only 168 hours a week. I also deleted error value categories such as No hours, Hours vary, and NIU (Not in Universe).

Moreover, I created a new variable named "experience" to capture the experience years of individuals by subtracting their education years and the number 6 from their age as follow:

$$\text{experience} = \text{age} - \text{educyears} - 6$$

This helps isolate actual work experience beyond formal education and childhood years before primary schools while standardizing the measure for labor market analysis. Furthermore, I only

chose people from 25 to 64 years old since this is the working-age population, actively participating in the labor force. Addition to the "experience" variable is "experience_squared" in which I square the original variable to capture any non-linear relationship between experience years and hourly wages to have a more accurate representation of how the effect of experience on wages may change over time or at different stages of people's career.

TABLE I: SUMMARY STATISTICS OF NUMERICAL VARIABLES USING STATA

Variable	Obs	Mean	Std. Dev.	Min	Max
hourwage	3,795	23.36277	11.21546	1.12	99.99
educyears	65,154	14.04388	2.457546	9	18
uhrsworkt	46,889	40.72074	10.01753	1	140
experience	65,154	23.79507	11.72468	1	49
experience~d	65,154	703.6715	593.1729	1	2401

b. Dummy variables

Firstly, for the variable sex, I dropped all the NIU (Not in Universe) and chose only male and female before downloading the data from CPS. Then on Stata, I re-coded it as 1 for male and 0 for female, so female becomes the base case. This is the coding process for dummy variables, creating a binary variable representing gender male. The coefficient associated with the "male" dummy variable then quantifies the average difference in the dependent variable - hourly wage - between males and females in the model.

I also created an interaction term between education years (educyears) and gender (male), which is male*educyears that multiply these two factors together. This newly created variable is included in my regression models to examine whether the relationship between education years and the outcome variable hourly wage differs between males and females. This is important in understanding how gender influences the impact of education on earnings.

Furthermore, when choosing the variable for citizenship status on CPS, I removed categories "Born in U.S. outlying," "Born abroad of American parents," "Naturalized citizen," and "NIU (Not in Universe)" data to focus on specific subgroups relevant to my research question, only picked people who were born inside the US and those who were born outside. Then I re-coded these two groups on Stata and made a new binary variable “usCitizen” with people who are US citizens coded as 1 and those who are not coded as 0.

Lastly, the use of race dummies and occupation dummies is essential for capturing the heterogeneous nature of the labor market. For the variable race, I dropped blank data and people with unspecified race from my case selection process in CPS. On Stata, I re-coded it as a new variable “Race” using dummy coding so I have four main groups: White as 1, Black as 2, Asian as 3, and others as 4. This approach enables the examination of how race influences the outcome variable by comparing the average effect of each racial group against a base group (White). It also helps to capture the diverse racial composition of the population and provides insights into potential disparities or differences in outcomes across racial categories.

For variable occupation, I used the same strategy as re-coding for race, to get new 11 “occupation” dummies from the variable “occ” that I got data from Stata. They help me understand how different job roles and industries contribute to wage differentials. Since there are so many occupations, I looked up for “Major Occupation Group Recodes” from the United States Census Bureau to divided them into 11 groups as follow:

1 - Management, business, and financial occupations

2 - Professional and related occupations

3 - Service occupations

4 - Sales and related occupations

5 - Office and administrative support occupations

6 - Farming, fishing, and forestry occupations

7 - Construction and extraction occupations

8 - Installation, maintenance, and repair occupations

9 - Production occupations

10 - Transportation and material moving occupations

11 - Armed Forces

After filtering all the data, I realize that CPS indeed comes with a lot of data limitations and challenges. One of them is the presence of outliers and missing data, which require extensive sorting and cleaning efforts, potentially reducing the sample size available for analysis. The number of observations reduces to 3,795 after this process. In addition, according to *CEPR data*, the CPS is not a Simple Random Sample (SRS) of U.S. households but rather employs a "multistage stratified sample" approach. This means that not everyone has an equal probability of being selected for the survey, leading to potential biases in the data. "As a result, the procedures that statistical packages usually use to calculate standard errors will produce estimates that are systematically lower than they should be," underestimating the true variability in the data and impacting the accuracy of estimates and hypothesis testing (*CEPR data*).

TABLE II: PERCENTAGES OF PEOPLE AGE 25-64 IN VARIOUS CATEGORIES

Group	Number of people	Percentages
Sex	65,154	
Male (1=male)	31,657	48.59
Female (0=female)	33,497	51.41
Race		
1 - White only	51,315	78.76
2 - Black only	7,820	12.00
3 - Asian only	3,211	4.93
4 - Others	2,808	4.31
Citizenship		
US Citizen (1=yes)	57,053	87.57
non US Citizen (0=no)	8,101	12.43
Occupation	51,427	
1 - Management, business, and financial occupations	10,423	20.27
2 - Professional and related occupations	13,487	26.23
3 - Service occupations	7,202	14.00
4 - Sales and related occupations	4,067	7.91
5 - Office and administrative support occupations	5,099	9.92
6 - Farming, fishing, and forestry occupations	357	0.69
7 - Construction and extraction occupations	2,977	5.79
8 - Installation, maintenance, and repair occupations	1,624	3.16
9 - Production occupations	2,725	5.30
10 - Transportation and material moving occupations	3,462	6.73
11 - Armed Forces	4	0.01

B. Presentation and Interpretation of Results

In this research, I aim to provide a comprehensive understanding of the drivers of Rate of Return (ROR) on education differentials between male and female, also influenced by other factors by analyzing five regression models progressively.

My research paper utilizes Data Generating Process (DGP) to build 5 models calculating the earnings - hourly wages and employs a linear regression framework. The outcome variable, hourly wage (Y), is determined by several independent variables (X) for the population of my study. These independent variables include education level, male gender (female becomes the base case), interaction term between education and gender, citizenship status, hours worked, years of experience, experience squared (to capture nonlinear effects), four race dummies (to account for racial differences), and eleven occupation dummies (to control for variations in job types). My last model transforms hourly wage to ln hourwage to interpret coefficients in terms of percentage changes in hourly wage rather than absolute changes:

$$\ln(W_i) = \ln(W_0) + \mathbf{X}_i\mathbf{B} + \varepsilon_i$$

My economic model is referenced from Mincer (1974) earnings function, where:

- $\ln(W_i)$ represents the natural logarithm of the hourly wage being calculated by the model
- $\ln(W_0)$ represents the intercept or base level of the model
- $\mathbf{X}_i$ is a vector representing all independent variables for observation i
- $\mathbf{B}$ is a vector of coefficients corresponding to the independent variables, analogous to the beta coefficients (β)
- ε_i represents the error term for observation i

TABLE III: OLS REGRESSION ESTIMATES OF THE EFFECTS OF EDUCATION ON HOURLY WAGE

	Model 1: HOURWAGE	Model 2: HOURWAGE	Model 3: HOURWAGE	Model 4: HOURWAGE	Model 5: Ln HOURWAGE
Intercept	-1.236 (1.310)	-4.397 (1.352)	-7.227 (1.699)	-21.79 (2.142)	1.883 (0.0783)
educyears	1.831 (0.103)	1.945 (0.104)	2.151 (0.133)	2.344 (0.140)	0.0605 (0.00463)
male (1 = yes)		3.342 (0.344)	9.384 (2.660)	10.68 (2.707)	0.289 (0.0849)
male*educyears			-0.451 (0.211)	-0.573 (0.215)	-0.0146 (0.00649)
usCitizen (1 = yes)				1.171 (0.491)	0.0232 (0.0185)
hours worked per week				0.116 (0.0209)	0.00592 (0.000810)
experience				0.504 (0.0636)	0.0174 (0.00226)
experience^2				-0.00752 (0.00126)	-0.000249 (0.0000447)
4 Race dummies	No	No	No	No	Yes
11 occupation dummies	No	No	No	No	Yes
N	3,795			3,601	
R Squared	0.115	0.137	0.139	0.179	0.274

- ***Model 1 - Basic Model: Hourly Wage on Education Years***

In the initial regression model, I regress hourly wage solely on education years. The coefficient for education years represents the average change in hourly wage associated with each additional year of education, holding all other variables constant. A positive coefficient indicates that higher levels of education are associated with higher hourly wages, with roughly

$1.83 ± $0.1 increase in hourly wages for each education year. This model has 3,795 observations, with the R-squared value of 0.115, indicating that approximately 11.5% of the variance in hourly wage is explained by the model.

- ***Model 2 - Gender Analysis: Hourly Wage on Education and Male Gender***

In this model, I introduce gender (coded as 1 for male and 0 for female) as an additional predictor alongside education years. The coefficient for male gender signifies the average difference in hourly wage between males and females, controlling for education. A positive coefficient indicates that, on average, males earn $3.34 ± $0.3 more per hour than females with the same level of education. The model explains approximately 13.7% of the variance in hourly wage (R-squared = 0.137).

- ***Model 3 - Interaction Effect: Hourly Wage on Education, Male Gender, and Interaction Term***

Adding an interaction term between education years and male gender allows me to assess if the relationship between education and hourly wage differs significantly between males and females. A significant interaction term indicates that the impact of education on wages varies between genders. An interesting result is, although males tend to earn around $9.38 ± $2.7 per hour more than females when holding education constant and before accounting for the interaction effect, each additional year of education, the hourly wage for males is $0.45 ± $0.2 lower than the corresponding increase for females. The following equation is used to calculate the predicted hourly wage for female (the base case):

$$\text{Predicted Hourly Wage}_F = -7.227 + 2.151*\text{educyears}$$

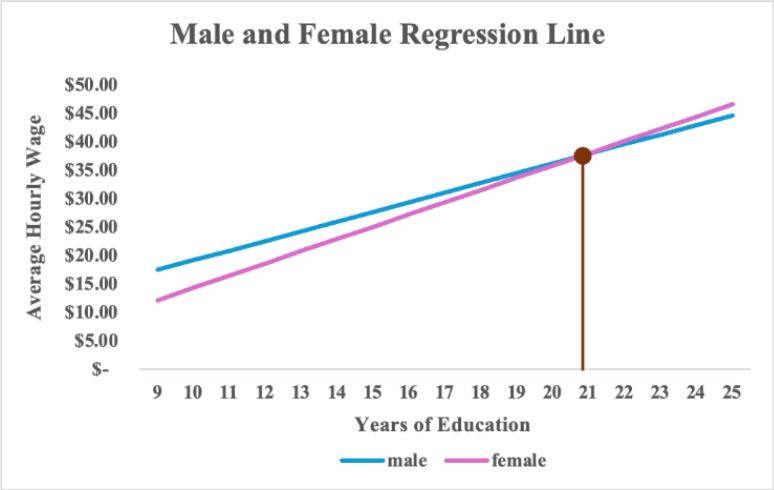
For calculating the predicted hourly wage for males, we cannot simply look at the coefficient for "male" anymore, since it is included in the interaction term male*educyears. Therefore, in the equation to calculate the wage for men, we must substitute a "1" for the "male" variable and also include the coefficient of the interaction term (Barreto, 2005):

$$\text{Predicted Hourly Wage}_M = -7.227 + 2.151*\text{educyears} + 9.384 - 0.451*\text{educyears}$$

FIGURE 1: GRAPHICAL INTERPRETATION OF THE INTERACTION TERM

Predicted Wage from Regression of Hourly Wage on Education, Male, and Male Education (Interaction Term)

educyears	male	female	M-F differential
9	$ 17.46	$ 12.13	$ 5.33
10	$ 19.16	$ 14.28	$ 4.87
11	$ 20.86	$ 16.43	$ 4.42
12	$ 22.56	$ 18.59	$ 3.97
13	$ 24.26	$ 20.74	$ 3.52
14	$ 25.96	$ 22.89	$ 3.07
15	$ 27.66	$ 25.04	$ 2.62
16	$ 29.36	$ 27.19	$ 2.17
17	$ 31.06	$ 29.34	$ 1.72
18	$ 32.76	$ 31.49	$ 1.27
19	$ 34.46	$ 33.64	$ 0.81
20	$ 36.16	$ 35.79	$ 0.36
21	$ 37.86	$ 37.94	$ (0.09)
22	$ 39.56	$ 40.10	$ (0.54)
23	$ 41.26	$ 42.25	$ (0.99)
24	$ 42.96	$ 44.40	$ (1.44)
25	$ 44.66	$ 46.55	$ (1.89)



Source: InteractionTerm_MaleEduc_Model3.xlsx

From this calculation, we can estimate that after 20.8 years of education (9.384 / 0.451), male and female will likely have the equal hourly wages. This is an unexpected finding revealing a reversal of the traditional assumption that males earn more than females with the same level of education. The result challenges conventional assumptions about gender-based earnings in higher education cohorts. Now the R-squared value increases to 0.139.

- ***Model 4 - Comprehensive Model: Hourly Wage on Education, Male Gender, Interaction Term, Citizenship, Hours Worked, Experience, Experience Squared***

Expanding the model to include citizenship status, hours worked per week, experience, and experience squared provides a more comprehensive analysis. The estimated coefficients for these variables show how each factor influences hourly wage, accounting for gender differences and the nonlinear effect of experience. The results show that holding other factors constant, US Citizens earn around $1.17 ± $0.5 per hour more than non-citizen, and each year increase in experience will lead to a higher hourly wage. Furthermore, each additional year of experience is associated with around $0.5 increase in hourly wage, on average, holding other factors constant. However, the effect of experience is moderated by the squared experience term. The negative estimated coefficient of experience squared (-0.00752) in my model 4 suggests that the positive effect of experience on hourly wage diminishes at higher levels of experience, following a concave (inverted U-shaped) pattern. The turning point (maximum) of the experience-wage profile can be calculated as:

maximum experience = - experience coefficient / (2 * experience squared coefficient).

With the result of 33.5 (-0.504 / (2 * -0.00752)), this means that the positive effect of experience on hourly wage increases until around 33.5 years of experience, after which the effect starts to decrease. The R-squared value for this model is 0.179.

- ***Model 5 - Log-Transformed Model: ln(Hourly Wage) on Education, Male Gender, Interaction Term, Citizenship, Hours Worked, Experience, Experience Squared, Race Dummies, Occupation Dummies***

In the final model, I use the natural logarithm of hourly wage to analyze percentage changes. Including race dummies and occupation dummies allows me to explore how race and occupation affect wage differentials after controlling for education, gender, and other variables.

From estimated coefficients results in table III, we can conclude that an additional year of education is associated with a 6.05% increase in hourly wage, on average, holding other factors such as gender, citizenship status, hours worked per week, experience, experience squared, race, and occupation constant. The robust standard error (SE) of 0.00463 indicates the precision of this estimate. Moreover, males have approximately 28.9% higher hourly wages than females, on average, before accounting for the interaction effect and holding other factors constant. However, when considering the coefficient of the interaction term, the impact of education on Rate of Return (ROR) on education (in terms of percentage increase in hourly wage) is still slightly weaker for males compared to females in this optimal model. For each additional year of education, the gap between hourly wage for men and women is closer by 1.64%, indicating that the change in hourly wage for each higher education level in males is 1.64% lower than the corresponding increase for females, holding other factors constant. This means after 19.8 years of schooling, they will have equal hourly wage. The very small robust SE of 0.00649 suggests that the estimated coefficient for this interaction is relatively precise and reliable.

Notably, being a US citizen is associated with a 2.32% higher hourly wage, on average, compared to non-citizens, holding other factors constant. The estimated coefficients of 4 Race dummies indicate that compared to White people, Black individuals have approximately 9.67% lower hourly wages, on average, holding all other factors (education, gender, citizenship, hours worked, experience, occupation) constant. Meanwhile, Asians and other racial groups have the hourly wage around 0.3% and 6.12%, respectively, lower than Whites. Considering occupational

choices, people in Professional and related occupations have the highest ROR on education, followed by the group of Management, business, and financial occupations. For this model, the R-squared value is the highest among these five, explaining approximately 27.4% of the variance in Rate of Return (ROR) on education (R-squared = 0.274).

Analyzing deeper into this optimal model, I examine its p-value to see if my independent variables' coefficients have a significant impact on the dependent variable ln hourwage.

TABLE IV: REGRESSION RESULTS FOR MODEL 5 USING STATA

Linear regression

Number of obs	=	3,601
F(19, 3581)	=	69.91
Prob > F	=	0.0000
R-squared	=	0.2744
Root MSE	=	.36146

ln_hourwage	Coef.	Robust Std. Err.	t	P>|t|	[95% Conf.	Interval]
educyears	.0605358	.0046305	13.07	0.000	.0514571	.0696146
male	.2889603	.0848555	3.41	0.001	.1225904	.4553303
maleeducyears	-.0145704	.0064862	-2.25	0.025	-.0272873	-.0018535
usCitizen	.0232025	.0185316	1.25	0.211	-.0131311	.0595361
uhrsworkt	.0059166	.0008102	7.30	0.000	.0043281	.0075052
experience	.0173786	.0022577	7.70	0.000	.012952	.0218051
experience_squared	-.0002493	.0000447	-5.57	0.000	-.000337	-.0001616
Race						
2	-.0966846	.0172191	-5.61	0.000	-.1304448	-.0629243
3	-.0031186	.0478956	-0.07	0.948	-.0970241	.0907868
4	-.0611792	.0288834	-2.12	0.034	-.1178088	-.0045495
occupation						
2	.0157706	.0285679	0.55	0.581	-.0402404	.0717817
3	-.3351918	.029356	-11.42	0.000	-.3927479	-.2776356
4	-.2872291	.0313937	-9.15	0.000	-.3487804	-.2256778
5	-.1882799	.02829	-6.66	0.000	-.2437461	-.1328138
6	-.3718392	.0682751	-5.45	0.000	-.5057013	-.2379772
7	-.0427724	.0326199	-1.31	0.190	-.1067279	.0211832
8	-.0087779	.0350053	-0.25	0.802	-.0774103	.0598544
9	-.1847626	.0293901	-6.29	0.000	-.2423856	-.1271395
10	-.2117865	.0301173	-7.03	0.000	-.2708354	-.1527377
_cons	1.882883	.0783347	24.04	0.000	1.729298	2.036468

Since the column P>|t| (the p-value associated with the t-statistic for particular coefficients, used for hypothesis testing and calculated by (coefficient estimate/standard error)) almost displays the

results less than 0.05, taking the coefficients of educyears, male, interaction term, hours worked per week (uhrsworkt), experience, experience squared, and Black people (coded as 2 in Race) as examples, this means the coefficients of these independent variables are statistically significant at a 95% confidence level. The p-value helps assess the significance of each variable in the regression model. In other words, there is strong evidence to reject the null hypothesis that these coefficients are equal to zero. Conversely, the p-value of the coefficients of citizenship status, Asian people (coded as 3), and some occupations shown in this column are higher than 0.05, meaning that these coefficients are not statistically significant. This indicates that these coefficients may not have a meaningful impact on the dependent variable ln hourwage in my regression model.

V. CONCLUSION

In conclusion, this research paper examines the crucial question of whether the Rate of Return (ROR) on education differs between men and women in the United States labor market. Through comprehensive regression analysis using data from the Integrated Public Use Microdata Series (IPUMS) CPS (Current Population Survey), March 2023 ASEC, interesting numbers were revealed. Overall, each additional year of education increases the income by 6.05%, and men tend to earn 28.9% more than women if we only consider gender factors. However, this gap is closed by 1.46% for each year of schooling, leading to females with 20 years of education and up will earn even higher than males. The results challenge traditional assumptions about the Gender Wage Gap, highlighting the need for further research in understanding the underlying mechanisms that contribute to these wage differentials and how they evolve over time. Additionally, the analysis revealed significant racial disparities in earnings, especially when

Black people earn 9.67% less than Whites, and varying ROR across different occupational choices.

My findings align closely with previous studies, particularly with Sorel and Shinners (2017), who found that for each additional year of education, wages increased by 12.6%. In contrast, my analysis reveals a 6.05% increase in income per year of education. This discrepancy may stem from different data sources or sample characteristics. However, both studies underscore the positive impact of education on income levels, supporting the principles of Human Capital Theory. Additionally, while my study highlights a 28.9% initial income gap between males and females, Sorel and Shinners (2017) discovered a more substantial disparity, with women earning 48.7% less than men when holding all other factors constant, and Bertrand, Goldin, and Katz (2010) even emphasized the gap of 56.2% for men and women in the ninetieth percentile. These findings suggest that while education may narrow the gender wage gap over time, significant disparities persist, necessitating continued efforts to address structural inequalities in the labor market. Kochhar from Pew Research Center (2023)'s findings further support my observations, reporting an 18% gender wage gap in 2022, with women earning approximately 82 cents for every dollar earned by men. These statistics underscore the ongoing challenges of gender-based income disparities and the need for comprehensive policies to promote gender equality in earnings.

Furthermore, my study's findings underscore several key economic implications. Firstly, the significant increase in income with each additional year of education highlights the economic value of investing in education. This suggests that policies and initiatives aimed at improving access to education and promoting lifelong learning can have substantial economic benefits by enhancing my earning potential and overall productivity. Secondly, the narrowing of the gender

wage gap with increased education levels indicates the potential for education to mitigate gender-based income disparities, contributing to greater economic equity and inclusivity. Additionally, the identification of significant racial disparities in earnings emphasizes the importance of addressing systemic inequalities within the labor market to ensure fair and equitable economic outcomes for all demographic groups. Overall, my study contributes valuable insights into the complex dynamics of income inequality, highlighting the role of education and gender in shaping earning outcomes. It underscores the economic importance of education in fostering economic growth, reducing income inequality, and promoting a more inclusive and prosperous society.

Future areas of exploration could include delving deeper into my dataset limitations and exploring alternative data sources that offer a more representative sample size and diverse population coverage. Given that the Current Population Survey (CPS) resulted in a reduced sample size of 3,795 observations after data sorting, future research could benefit from utilizing datasets that provide a larger and more comprehensive sample to better capture the complexities of income differentials across various demographics. Moreover, further research could explore the impact of ethnicity on hourly wage differentials by including a specific variable such as "Hispan," allowing for a deeper analysis of income disparities within diverse demographic groups. Additionally, investigating the impact of specific industries or sectors on Rates of Return (ROR) could offer valuable insights into how economic factors influence earnings inequalities, especially between people with STEM degrees and those who do not. Moreover, exploring the role of workplace policies, discrimination, and other structural factors in shaping wage differentials could further enhance our understanding of the mechanisms driving income gaps. Longitudinal data analysis could also be employed to track changes in wage differentials and

career trajectories over time, providing a more nuanced understanding of income dynamics and the factors influencing economic outcomes.

VI. REFERENCES

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